Snowflake 7TS Fl 29-31 - CW pump controls - Siemens call 10/30/2026 10:00

Siemens: Alan Parrish (Ops Mgr), Brendan Jackson (PM), Brian Greda, Rachael Werner. NW: Ruslan, Christino, Andrei. Tabs 01-08 refer to the files in this folder / bookmarks in the combined pack.

Goal of the call

• Agree the pump control scope: no separate panel, no $50k/floor. Pumps do duty/standby themselves; Siemens provides the points already in its proposal.

• Get a date for ATC submittal R1 (CRAC units on site mid-November, start-up after).

• Close the 9/28 items list (below).

Our position in four lines

• The design never had a separate panel: EOR specifies pumps with **integrated VFD and built-in pump system controller** (M-101 2.07.D items 5, 15, 17) and schedules them as **two pumps primary-standby** (M-401). No revision, no bulletin on this (tab 03).

• Sequence 3b asks for lead/lag *sequencing*: 'one pump running and the other as stand-by'. That is the Grundfos multipump 'Alternating operation': duty/standby, standby starts on alarm, no external controller (IOM 7.49 p.61, Dolphin guide p.3 - tabs 05, 06).

• Siemens had the approved pump submittal and the Grundfos manual on **6/29** (our email proposing multipump instead of a panel, resent 7/9). No objection; R0 came 7/22 still drawn with a panel and VFDs (tab 07).

• Siemens' own proposal p.5 already lists the field points (flow switches, leak detectors to BB BMS and autodialer, start/stop and status, DP transmitter, AC interlock). If a panel is still wanted: Alt 3 is **$2,800 per panel** (tab 01).

Answers to Alan (emails 10/28 and 10/29)

Alan says	Answer	Where
No panel, something in the VFDs does the rotation - so no BMS interaction for rotation?	Correct. Pumps are linked by GENIbus and run Alternating operation. No BMS logic for rotation.	05 p.61; 06 p.3
No alarm point in the documents / proposal; '100% standalone, no connection to the BMS at all'	'No connection to base building BMS' is about pump control. Alarms ARE in scope: leak detectors 'to alarm at BB BMS and Tenant AutoDialer', flow switches, start/stop + status (p.5); Excl. 6 'except Leak Detection'; seq 3c/3d local A/V alarm + landlord BMS - copied on their own 300C; their 300A already wires flow and leak alarms to DXRs.	01 p.5, p.8; 02 marks G, H, J, K; 03 M-102
Did the VFD vendor confirm the drives act as the lead/lag panel?	Yes - Dolphin 9/28: duty/standby guide, 'Standard startup/programming can be used', 'Contractor needs to run the interconnecting wiring', 'One pressure sensor is required per MLE motor'. Start-up scope confirmation requested.	06; 07
Pump panels should come with the mechanical package (Alan 10/27)	On this job the EOR specified the controller inside the pump; nothing else is scheduled. M-102 ATC spec: control panels/power are controls contractor's work.	03

Scope split we want to agree (per floor, x3)

Item	Who	Basis
Pumps, check valves, drip pans, piping, DP taps; install DP transmitters and flow switches (pipe devices)	NW	Siemens Excl. 18 'installation of pipe mounted devices'
GENIbus cable between the two pumps (A-Y-B, 16 ga shielded)	NW / EC (low voltage) - agree on call	Dolphin: 'contractor'
Pump multipump setup, sensor setup, I/O settings at start-up	Dolphin (confirming)	Barry 9/28
CRAC run (enable) contact -> DI1 of both pumps, jumper 2-6 removed	Siemens	p.5 'Hardwire interlock to AC unit'
Flow switches FS-1/FS-2: furnish and wire; contact to DXR (alarm) + to that pump's DI2 (External fault)	Siemens	p.5 'Flow Switches - furnish and wire'

Item	Who	Basis
Leak detectors LD (pump pans): to DXR, BB BMS, autodialer + contact to that pump's DI3	Siemens	p.5
DP transmitters: 2 per floor (one per pump) wired to AI1 4-20 mA	Siemens	Grundfos / Barry; IOM 'running pump sensor'
Pump status / alarm (relay 1 Alarm, OC1 or relay 2 Running) to DXR / autodialer	Siemens	p.5 'Start/stop and Status'
Pump isolation valves V-1/V-2, one relay per valve (leak on one pump closes only that pump)	Siemens	p.5; seq 3d
Local audio/visual alarm (horn/strobe) for loss of flow and leak	Siemens	seq 3c/3d on their 300C
1 h in 4 h run timer (3f)	Met if CRAC runs 24/7 (4a IDF); else DXR timer contact in parallel on DI1	seq 3f / 4a
Delete from R0: lead/lag panel box, VFD panels, VFD run/speed wiring, CS-1/CS-2 optional	Siemens R1	9/28 email

Questions to ask Siemens

• With no panel and no BMS rotation, what exactly is left in the '$50k per floor'? Itemize it.

• Do you agree the field points on proposal p.5 stay in base scope (flow switches, leak detectors, valves, status, interlock, DP)?

• How will you provide the local audio/visual alarm (3c/3d) and the 3f timer - both are in your own 300C sequence but not on any sheet?

• Flow switches: proposal says 'furnish and wire', R0 shows 'FBO' and not in BOM. Confirm you furnish.

• V-1/V-2: missing from the valve schedule; both on RE-1. Will R1 show one relay per valve?

• CRAC valves (100A): open with unit, close when off, 4 h daily flush (4h); 5-minute delay (4i/4j); separate condensate alarm.

• Which contact is the 'unit status' RE-x-1 - fan/enable or compressor? Must be enable/fan, or CRAC and pumps never start (flow switch 4k).

• Who powers the valve actuators and relays at start-up - DXR 24 V as on 300A?

• Date for ATC submittal R1, and who from Siemens attends start-up with Dolphin.

• If you still want a panel: Alt 3 at $2,800 per panel - what would it do that the pumps do not?

Keep in mind

• DP transmitters: 2 per floor needed (R0 has 1). Do not raise the extra cost - wait for Siemens to bring it up.

• Alternation: Grundfos recommends 'energy'; 'time' needs FM310/FM311 per current IOM p.63 (our pumps FM 300 per submittal I/O). Spec says 150 h - minor deviation; note for the EOR if asked.

• Do not use the pump DI 'Flow switch' function (that is low-flow stop, needs a diaphragm tank). Low-flow stop = Not active.

• Bypass HD-053 still open (Siemens RFI). Ask status.

• Fallback if Siemens insists on a panel: (1) Alt 3 at $2,800/panel ($8,400); (2) NW furnishes a Tekmar 132 per pair (~$800-1,100; tab 08) and Siemens installs, powers and wires it per p.5 'Install and power manufacturer provided Lead / Lag panel'. Mention only if needed.

Still open from our 9/28 email

• Flow switches in BOM

• V-1/V-2 in valve schedule

• One relay per pump valve (3d)

• 3c/3d/3f alarm and timer

• 300B typos (CWP-29 labels on fl. 31, LD-8 twice, DPTE-1)

• CRAC valve logic 4h; 4i/4j delay; condensate alarm

- DP range 0-15 vs 0-25 psi
- Sheet 100 relay table all WCCU-29-1

Tab 02 legend - marks on Siemens ATC R0

Mark	Sheet	What it shows
A	300	'LEAD LAG PUMP CONTROLLER' box = Siemens' own RIBU1C relays RE-1..5 (BOM 15) - their relays were to sit at the by-others panel
B	300	Note 2 'VFD mounted in ceiling near pump set' - the VFD is integral to the pump motor
C	300	Flow switches FS-1/FS-2 marked 'FBO'; proposal p.5 says furnish and wire
D	300	One DP transmitter per floor, 0/15 psi (BOM says 25 psi); Grundfos needs one per pump
E (red, 300A/300B)	300A/B	'CWP-xx-1&2 LEAD LAG PANEL' by others - not in BOM, no model, no power; terminal numbers 2/6, 4/6, 12/18, 10/20 are the Grundfos CRE motor terminals
F (blue)	300A/B	'VFD panels' - do not exist; 300B labels the 31st floor as CWP-29
G (green)	300A/B	Flow alarms RE-2-1 / RE-4-1 already wired to Siemens DXRs
H (green)	300A/B	Leak detectors LD-1..4 already wired to Siemens DXRs / autodialer
I	300C	BOM: no flow switches, no panel, no horn/strobe
J, K, L	300C	Their sequence copies M-102 3c (local A/V alarm), 3d (leak local + landlord BMS), 3f (timer)

Tab 07 - Barry (Dolphin) 9/28 and our 6/29 email to Siemens

Date / from	Text (verbatim)
9/28 8:09 Barry	Wiring diagram was included in the submittal. I am attaching it again. (generic 'control by BMS' diagram)
9/28 10:50 Barry	I am not sending you the generic drawing but that's how the pump was designed, specified, approved & released. Our startup tech will also start the pump based on this drawing only. Any advanced programming or pump setup will be by others. I will try to get answers to your comments below from the factory.
9/28 3:31 pm Barry	Duty/standby instructions are attached. -Standard startup/programming can be used. -Contractor needs to run the interconnecting wiring. -One pressure sensor is required per MLE motor.
Open with Barry	Draft 10/27 (not sent): confirm start-up does multipump + DP sensor + I/O settings on all 3 pairs, price if extra, notice to schedule; panel option price. Weak spot: '10:50' line says advanced programming by others - get the confirmation in writing.
6/29 Ruslan to Brendan + Brian	Since each floor operates as an independent system serving only one data room AC unit, we can use the native multi-pump functionality built into the Grundfos CRE drives. See attached Manual ... Section 7.49 ... By linking the two pump heads directly via their GENIbus communication terminals, they will automatically handle all duty rotation, staging, and standby redundancy on their own. With this setup and since pumps not going to base building BMS, i think lead/lag panel not necessary here. (Attached: Grundfos IOM and approved H-02.00.) Resent 7/9 'please confirm'. No reply on this point; R0 issued 7/22.